T3 intermediate-EFT one-loop RGE term comparison: EFT_1 after F

This stage has active field content SM$+S_1+S_2$. The report combines the renormalisable RGBeta running with the dimension-five $\psi^2\phi^2$ Wilson-coefficient running generated after integrating out F. Throughout, $16\pi^2\,\mu\,dX/d\mu = \beta_X^{(1)}$. For the Wilson sector only the tree-generated stage-1 coefficient $C^{(0)}$ is inserted into the one-loop anomalous dimension; the one-loop threshold piece is retained as a boundary term and is not run again at this order.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W^A_\mu$
g_3	$D_\mu \supset i g_3 t^A G^A_\mu$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5}\supset R_W\lambda_5C_{LLS_1S_2}$

Renormalisable EFT1 couplings

$\beta^{(1)}_{g_1}$

Quantity: g_1

$$16\pi^2\,\mu\frac{dg_1}{d\mu}=\beta^{(1)}_{g_1}$$

Column	Representative term structure
T1	$\frac{47{g_1}^3}{6}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{47{g_1}^3}{6}$

$\beta_{g_2}^{(1)}$

Quantity: g_2

$$16\pi^2\,\mu\,\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$$

Column	Representative term structure
T1	$-\frac{5{g_2}^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5{g_2}^3}{2}$

$\beta_{g_3}^{(1)}$

Quantity: g_3

$$16\pi^2\,\mu\,\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$$

Column	Representative term structure
T1	$-7{g_3}^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7{g_3}^3$

$\beta^{(1)}_{Y_u}$

Quantity: Y_u

$$16\pi^2\,\mu\,\frac{dY_u}{d\mu} = \beta^{(1)}_{Y_u}$$

Terms 1 of 3

Column	Representative term structure
T1	$3\,\mathrm{Tr}\Big(Y_d\ \left(Y_d\right)^{\dagger}\Big)\ \left(Y_u\right)_{ij}$
T2	$+\,\mathrm{Tr}\Big(Y_e\ \left(Y_e\right)^{\dagger}\Big)\ \left(Y_u\right)_{ij}$
T3	$+3\,\mathrm{Tr}\Big(Y_u\ \left(Y_u\right)^{\dagger}\Big)\ \left(Y_u\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\ \left(Y_d\right)^{\dagger}\Big)\ \left(Y_u\right)_{ij}$	$+\,\mathrm{Tr}\Big(Y_e\ \left(Y_e\right)^{\dagger}\Big)\ \left(Y_u\right)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\ \left(Y_u\right)^{\dagger}\Big)\ \left(Y_u\right)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\,(Y_u)_{ij}$
T3	$-8{g_3}^2\,(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_u\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_u\right)_{ij})$	$-\frac{9}{4}{g_2}^2\,(Y_u)_{ij}$	$-8{g_3}^2\,(Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}{g_1}^2\,(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}{g_1}^2\,(Y_u)_{ij}$

$\beta^{(1)}_{Y_d}$

Quantity: Y_d

$$16\pi^2\,\mu\,\frac{dY_d}{d\mu} = \beta^{(1)}_{Y_d}$$

Terms 1 of 3

Column	Representative term structure
T1	$\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$
T2	$+3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$
T3	$+3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij}$

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$
T2	$-\frac{9}{4}{g_2}^2\,(Y_d)_{ij}$
T3	$-8{g_3}^2\,(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}(\left(Y_d\ (Y_d)^\dagger\ Y_d\right)_{ij} - \left(Y_u\ (Y_u)^\dagger\ Y_d\right)_{ij})$	$-\frac{9}{4}{g_2}^2\,(Y_d)_{ij}$	$-8{g_3}^2\,(Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12} {g_1}^2 \left(Y_d\right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12} {g_1}^2 \left(Y_d\right)_{ij}$

$\beta^{(1)}_{Y_e}$

Quantity: Y_e

$$16\pi^2\,\mu\,\frac{dY_e}{d\mu} = \beta^{(1)}_{Y_e}$$

Terms 1 of 2

Column	Representative term structure
T1	$3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij}$
T2	$+3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij}$
T3	$+\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij}$	$+3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij}$	$+\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 2

Column	Representative term structure
T1	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$
T2	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$
T3	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}\left(Y_e\ (Y_e)^\dagger\ Y_e\right)_{ij}$	$-\frac{9}{4}{g_2}^2\,(Y_e)_{ij}$	$-\frac{15}{4}{g_1}^2\,(Y_e)_{ij}$

$\beta^{(1)}_{m_1^2}$

Quantity: m_1^2

$$16\pi^2\,\mu\frac{dm_1^2}{d\mu} = \beta^{(1)}_{m_1^2}$$

Column	Representative term structure
T1	$6\lambda^{(1)}_{12}m_2^2$
T2	$+4\lambda^{(1)}_{S_1}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$6\lambda^{(1)}_{12}m_2^2$	$+4\lambda^{(1)}_{S_1}m_1^2$

$\beta^{(1)}_{m_2^2}$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu}=\beta^{(1)}_{m_2^2}$$

Terms 1 of 2

Column	Representative term structure
T1	$-12{g_2}^2m_2^2$
T2	$-6{g_1}^2m_2^2$
T3	$+2\lambda^{(1)}_{12}m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12{g_2}^2m_2^2$	$-6{g_1}^2m_2^2$	$+2\lambda^{(1)}_{12}m_1^2$

Quantity: m_2^2

Terms 2 of 2

Column	Representative term structure
T1	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+m_2^2(8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$

$\beta^{(1)}_{\lambda_1}$

Quantity: λ_1

$$16\pi^2\,\mu\,\frac{d\lambda_1}{d\mu} = \beta^{(1)}_{\lambda_1}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T2	$-12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$
T3	$-4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$-12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big)$	$-4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big)$

Quantity: λ_1

Terms 2 of 5

Column	Representative term structure
T1	$-12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$
T2	$+4\lambda_5\,(\lambda_5)^*$
T3	$+\frac{9{g_2}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big)$	$+4\lambda_5\,(\lambda_5)^*$	$+\frac{9{g_2}^4}{4}$

Quantity: λ_1

Terms 3 of 5

Column	Representative term structure
T1	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$
T2	$-9{g_2}^2\lambda_1$
T3	$+\frac{3{g_1}^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+{g_1}^2\left(\frac{3{g_2}^2}{2}-3\lambda_1\right)$	$-9{g_2}^2\lambda_1$	$+\frac{3{g_1}^4}{4}$

Quantity: λ_1

Terms 4 of 5

Column	Representative term structure
T1	$+12{\lambda_1}^2$
T2	$+2{\lambda^{(1)}_{HS_1}}^2$
T3	$+6{\lambda^{(1)}_{HS_2}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12{\lambda_1}^2$	$+2{\lambda^{(1)}_{HS_1}}^2$	$+6{\lambda^{(1)}_{HS_2}}^2$

Quantity: λ_1

Terms 5 of 5

Column	Representative term structure
T1	$+3\lambda_{HS_2}^{(A)\ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_2}^{(A)\ 2}$

$$\beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

Quantity: $\lambda^{(1)}_{S_1}$

$$16\pi^2\,\mu\,\frac{d\lambda^{(1)}_{S_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{S_1}}$$

Column	Representative term structure
T1	$6{\lambda^{(1)}_{12}}^2$
T2	$+4{\lambda^{(1)}_{HS_1}}^2$
T3	$+10{\lambda^{(1)}_{S_1}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$6{\lambda^{(1)}_{12}}^2$	$+4{\lambda^{(1)}_{HS_1}}^2$	$+10{\lambda^{(1)}_{S_1}}^2$

$\beta^{(1)}_{\lambda^{(1)}_{S_2}}$

Quantity: $\lambda^{(1)}_{S_2}$

$$16\pi^2\,\mu\frac{d\lambda^{(1)}_{S_2}}{d\mu}=\beta^{(1)}_{\lambda^{(1)}_{S_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$21{g_2}^4$
T2	$+{g_1}^2\left(12{g_2}^2-12\lambda^{(1)}_{S_2}-6\lambda^{(A)}_{S_2}\right)$
T3	$+{g_2}^2(-24\lambda^{(1)}_{S_2}-12\lambda^{(A)}_{S_2})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$21{g_2}^4$	$+{g_1}^2\left(12{g_2}^2-12\lambda^{(1)}_{S_2}-6\lambda^{(A)}_{S_2}\right)$	$+{g_2}^2(-24\lambda^{(1)}_{S_2}-12\lambda^{(A)}_{S_2})$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$+12{g_1}^4$
T2	$+2{\lambda_{12}^{(1)}}^2$
T3	$+4{\lambda_{HS_2}^{(1)}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12{g_1}^4$	$+2{\lambda_{12}^{(1)}}^2$	$+4{\lambda_{HS_2}^{(1)}}^2$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 4

Column	Representative term structure
T1	$+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$
T2	$+14\lambda_{S_2}^{(1)\,2}$
T3	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{HS_2}^{(A)\ 2}}{2}$	$+14\lambda_{S_2}^{(1)\,2}$	$+14\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 4

Column	Representative term structure
T1	$+26\lambda_{S_2}^{(A)\,2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{(A)\,2}$

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

Quantity: $\lambda^{(1)}_{HS_1}$

$$16\pi^2\,\mu\,\frac{d\lambda^{(1)}_{HS_1}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_1}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_1}\,(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$
T2	$+12\lambda_5\ (\lambda_5)^*$
T3	$-\frac{9{g_2}^2\lambda^{(1)}_{HS_1}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda^{(1)}_{HS_1}\,(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big))$	$+12\lambda_5\ (\lambda_5)^*$	$-\frac{9{g_2}^2\lambda^{(1)}_{HS_1}}{2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$
T2	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T3	$+6\lambda_1\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}{}^{2}$
T2	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)}{}^{2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$

$$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Quantity: $\lambda^{(1)}_{HS_2}$

$$16\pi^2\,\mu\,\frac{d\lambda^{(1)}_{HS_2}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{HS_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^{\dagger}\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^{\dagger}\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^{\dagger}\Big) + 4\lambda^{(A)}_{S_2})$
T2	$+4\lambda_5\ (\lambda_5)^*$
T3	$+6{g_2}^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^{\dagger}\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^{\dagger}\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^{\dagger}\Big) + 4\lambda^{(A)}_{S_2})$	$+4\lambda_5\ (\lambda_5)^*$	$+6{g_2}^4$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 4

Column	Representative term structure
T1	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$
T2	$+3{g_1}^4$
T3	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2}$	$+3{g_1}^4$	$-\frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2}$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$
T2	$+6\lambda_1\lambda^{(1)}_{HS_2}$
T3	$+4\lambda^{(1)}_{HS_2}{}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	$+6\lambda_1\lambda^{(1)}_{HS_2}$	$+4\lambda^{(1)}_{HS_2}{}^2$

Quantity: $\lambda^{(1)}_{HS_2}$

Terms 4 of 4

Column	Representative term structure
T1	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$
T2	$+6{\lambda^{(A)}_{HS_2}}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	$+6{\lambda^{(A)}_{HS_2}}^2$

$$\beta^{(1)}_{\lambda^{(1)}_{12}}$$

Quantity: $\lambda^{(1)}_{12}$

$$16\pi^2\,\mu\,\frac{d\lambda^{(1)}_{12}}{d\mu} = \beta^{(1)}_{\lambda^{(1)}_{12}}$$

Terms 1 of 3

Column	Representative term structure
T1	$4\lambda_5\,(\lambda_5)^*$
T2	$-12{g_2}^2\lambda^{(1)}_{12}$
T3	$-6{g_1}^2\lambda^{(1)}_{12}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$4\lambda_5\,(\lambda_5)^*$	$-12{g_2}^2\lambda^{(1)}_{12}$	$-6{g_1}^2\lambda^{(1)}_{12}$

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 3

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}{}^2$
T2	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}{}^2$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$
T2	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

$\beta_{\lambda_5}^{(1)}$

Quantity: λ_5

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu} = \beta_{\lambda_5}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda_5\,(6\,\mathrm{Tr}\Big(Y_d\ \ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ \ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ \ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)})$
T2	$-\frac{21{g_2}^2\lambda_5}{2}$
T3	$-\frac{9{g_1}^2\lambda_5}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_5(6\,\mathrm{Tr}\Big(Y_d\ \ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ \ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ \ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)})$	$-\frac{21{g_2}^2\lambda_5}{2}$	$-\frac{9{g_1}^2\lambda_5}{2}$

Quantity: λ_5

Terms 2 of 3

Column	Representative term structure
T1	$+2\lambda^{(1)}_{12}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda^{(1)}_{HS_1}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda^{(1)}_{12}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda^{(1)}_{HS_1}\lambda_5$

Quantity: λ_5

Terms 3 of 3

Column	Representative term structure
T1	$+4\lambda^{(1)}_{HS_2}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda^{(1)}_{HS_2}\lambda_5$

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

Quantity: $\lambda^{(A)}_{HS_2}$

$$16\pi^2\,\mu\,\frac{d\lambda^{(A)}_{HS_2}}{d\mu}=\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$$

Terms 1 of 3

Column	Representative term structure
T1	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\Big(Y_d\ \ (Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\ \ (Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\ \ (Y_u)^\dagger\Big)+16\lambda^{(A)}_{S_2})$
T2	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3	$+{g_1}^2\left(4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}\Big(Y_d\ \ (Y_d)^\dagger\Big)+2\,\mathrm{Tr}\Big(Y_e\ \ (Y_e)^\dagger\Big)+6\,\mathrm{Tr}\Big(Y_u\ \ (Y_u)^\dagger\Big)+16\lambda^{(A)}_{S_2})$	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$	$+{g_1}^2\left(4{g_2}^2-\frac{15\lambda^{(A)}_{HS_2}}{2}\right)$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 3

Column	Representative term structure
T1	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$
T2	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$

Quantity: $\lambda^{(A)}_{HS_2}$

Terms 3 of 3

Column	Representative term structure
T1	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$

$$\beta^{(1)}_{\lambda^{(A)}_{S_2}}$$

Quantity: $\lambda^{(A)}_{S_2}$

$$16\pi^2\,\mu\,\frac{d\lambda^{(A)}_{S_2}}{d\mu} = \beta^{(1)}_{\lambda^{(A)}_{S_2}}$$

Terms 1 of 4

Column	Representative term structure
T1	$\frac{3{g_2}^4}{4}$
T2	$+{g_1}^2\left(9{g_2}^2 - \frac{3\lambda^{(1)}_{S_2}}{2} - 12\lambda^{(A)}_{S_2}\right)$
T3	$+{g_2}^2(-3\lambda^{(1)}_{S_2} - 24\lambda^{(A)}_{S_2})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3{g_2}^4}{4}$	$+{g_1}^2\left(9{g_2}^2 - \frac{3\lambda^{(1)}_{S_2}}{2} - 12\lambda^{(A)}_{S_2}\right)$	$+{g_2}^2(-3\lambda^{(1)}_{S_2} - 24\lambda^{(A)}_{S_2})$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 4

Column	Representative term structure
T1	$+\frac{3{g_1}^4}{2}$
T2	$+\frac{{\lambda_{12}^{(1)}}^2}{4}$
T3	$+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3{g_1}^4}{2}$	$+\frac{{\lambda_{12}^{(1)}}^2}{4}$	$+\frac{{\lambda_{HS_2}^{(1)}}^2}{2}$

Quantity: $\lambda^{(A)}_{S_2}$

Terms 3 of 4

Column	Representative term structure
T1	${+3\lambda^{(A)}_{HS_2}}^{2}$
T2	$+\frac{7{\lambda^{(1)}_{S_2}}^{2}}{4}$
T3	$+13\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	${+3\lambda^{(A)}_{HS_2}}^{2}$	$+\frac{7{\lambda^{(1)}_{S_2}}^{2}}{4}$	$+13\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 4

Column	Representative term structure
T1	$+22\lambda_{S_2}^{(A)\,2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{(A)\,2}$

Dimension-five $\psi^2\phi^2$ Wilson coefficients

The Wilson tensor is reported in the real-scalar component basis C_{ijab}. Each component with a nonzero one-loop beta function has its own table. Components absent at the matching boundary but generated by operator mixing are included automatically.

$\beta^{(1)}_{C_{1133}}$

Quantity: $1, 1, 3, 3$

$$16\pi^2\,\mu\,\frac{dC_{1133}}{d\mu} = \beta^{(1)}_{C_{1133}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3} lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3} lambda T_3 \overline{y_1 y_2}}{3MF}$

$$\beta^{(1)}_{C_{1134}}$$

Quantity: 1, 1, 3, 4

$$16\pi^2\,\mu\frac{dC_{1134}}{d\mu}=\beta^{(1)}_{C_{1134}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}ilambdaT_3\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{1144}}$

Quantity: 1, 1, 4, 4

$$16\pi^2\,\mu\,\frac{dC_{1144}}{d\mu} = \beta^{(1)}_{C_{1144}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3} lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3} lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{1157}}$

Quantity: $1, 1, 5, 7$

$$16\pi^2\,\mu\,\frac{dC_{1157}}{d\mu} = \beta^{(1)}_{C_{1157}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 1, 5, 7

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye \overline{y_1} \overline{y_2} \overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ye \overline{y_1} \overline{y_2} \overline{ye}}{12MF}$

$\beta^{(1)}_{C_{1158}}$

Quantity: $1, 1, 5, 8$

$$16\pi^2\,\mu\,\frac{dC_{1158}}{d\mu} = \beta^{(1)}_{C_{1158}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 1, 5, 8

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} iye\overline{y_1}\overline{y_2}\overline{ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} iye\overline{y_1}\overline{y_2}\overline{ye}}{12MF}$

$\beta^{(1)}_{C_{11511}}$

Quantity: $1, 1, 5, 11$

$$16\pi^2\,\mu\,\frac{dC_{11511}}{d\mu} = \beta^{(1)}_{C_{11511}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 1, 5, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}} g_2^2 \overline{y_1 y_2}}{3MF}$	$\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{12MF}$

$\beta^{(1)}_{C_{11512}}$

Quantity: 1, 1, 5, 12

$$16\pi^2\,\mu\,\frac{dC_{11512}}{d\mu} = \beta^{(1)}_{C_{11512}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{24MF}$

Quantity: 1, 1, 5, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\, ye\overline{y_1}\overline{y_2}\overline{ye}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, ig_2^2\overline{y_1}\overline{y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}\, ye\overline{y_1}\overline{y_2}\overline{ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, ig_2^2\overline{y_1}\overline{y_2}}{3MF}$

$\beta^{(1)}_{C_{1167}}$

Quantity: $1, 1, 6, 7$

$$16\pi^2\,\mu\,\frac{dC_{1167}}{d\mu} = \beta^{(1)}_{C_{1167}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$	$\frac{11 \cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 1, 6, 7

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{12MF}$

$\beta^{(1)}_{C_{1168}}$

Quantity: $1, 1, 6, 8$

$$16\pi^2\,\mu\,\frac{dC_{1168}}{d\mu} = \beta^{(1)}_{C_{1168}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 1, 6, 8

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$

$\beta^{(1)}_{C_{11611}}$

Quantity: $1, 1, 6, 11$

$$16\pi^2\,\mu\,\frac{dC_{11611}}{d\mu} = \beta^{(1)}_{C_{11611}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 1, 6, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{11612}}$

Quantity: 1, 1, 6, 12

$$16\pi^2\,\mu\,\frac{dC_{11612}}{d\mu} = \beta^{(1)}_{C_{11612}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 1, 6, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\, iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{3MF}$	$\frac{3^{\frac{3}{4}}\, iye\overline{y_1 y_2 ye}}{12MF}$

$$\beta^{(1)}_{C_{1213}}$$

Quantity: 1, 2, 1, 3

$$16\pi^2\,\mu\frac{dC_{1213}}{d\mu}=\beta^{(1)}_{C_{1213}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_{3}\overline{y_{1}y_{2}}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}lambdaT_{3}\overline{y_{1}y_{2}}}{3MF}$

$$\beta^{(1)}_{C_{1214}}$$

Quantity: 1, 2, 1, 4

$$16\pi^2\,\mu\,\frac{dC_{1214}}{d\mu} = \beta^{(1)}_{C_{1214}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3}ilambdaT_3\overline{y_1y_2}}{3MF}$

$\beta^{(1)}_{C_{1223}}$

Quantity: 1, 2, 2, 3

$$16\pi^2\,\mu\,\frac{dC_{1223}}{d\mu} = \beta^{(1)}_{C_{1223}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3} i l a m b d a T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{3} i l a m b d a T_3 \overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{1224}}$

Quantity: 1, 2, 2, 4

$$16\pi^2\,\mu\frac{dC_{1224}}{d\mu} = \beta^{(1)}_{C_{1224}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3} lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{\sqrt{3} lambda T_3 \overline{y_1 y_2}}{3MF}$

$$\beta^{(1)}_{C_{1259}}$$

Quantity: $1, 2, 5, 9$

$$16\pi^2\,\mu\,\frac{dC_{1259}}{d\mu} = \beta^{(1)}_{C_{1259}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{24MF}$

Quantity: 1, 2, 5, 9

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2\overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} g_2^2\overline{y_1 y_2}}{6MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1 y_2}}{6MF}$

$$\beta^{(1)}_{C_{12510}}$$

Quantity: 1, 2, 5, 10

$$16\pi^2\,\mu\,\frac{dC_{12510}}{d\mu} = \beta^{(1)}_{C_{12510}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 1, 2, 5, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}} i ye \overline{y_1 y_2 ye}}{MF}$
T3	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{25 \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}} i ye \overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$

$\beta^{(1)}_{C_{1269}}$

Quantity: $1, 2, 6, 9$

$$16\pi^2\,\mu\,\frac{dC_{1269}}{d\mu} = \beta^{(1)}_{C_{1269}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{25\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{6MF}$

Quantity: 1, 2, 6, 9

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}} i ye \overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} i g_1^2 \overline{y_1 y_2}}{8MF}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}} i ye \overline{y_1 y_2 ye}}{12MF}$

$\beta^{(1)}_{C_{12610}}$

Quantity: 1, 2, 6, 10

$$16\pi^2\,\mu\,\frac{dC_{12610}}{d\mu} = \beta^{(1)}_{C_{12610}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$	$\frac{25\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{24MF}$

Quantity: 1, 2, 6, 10

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\, ye\overline{y_1 y_2 ye}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{MF}$
T3	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}\, ye\overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{6MF}$	$-\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, ig_2^2\overline{y_1 y_2}}{6MF}$

$\beta^{(1)}_{C_{2211}}$

Quantity: 2, 2, 1, 1

$$16\pi^2\,\mu\,\frac{dC_{2211}}{d\mu}=\beta^{(1)}_{C_{2211}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3}lambdaT_3\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3}lambdaT_3\overline{y_1y_2}}{3MF}$

$$\beta^{(1)}_{C_{2212}}$$

Quantity: 2, 2, 1, 2

$$16\pi^2\,\mu\,\frac{dC_{2212}}{d\mu} = \beta^{(1)}_{C_{2212}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3} ilambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{2\sqrt{3} ilambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{2222}}$

Quantity: 2, 2, 2, 2

$$16\pi^2\,\mu\,\frac{dC_{2222}}{d\mu} = \beta^{(1)}_{C_{2222}}$$

Column	Representative term structure
T1	$\frac{\sqrt{3} lambda T_3 \overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{3} lambda T_3 \overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{2257}}$

Quantity: $2, 2, 5, 7$

$$16\pi^2\,\mu\,\frac{dC_{2257}}{d\mu} = \beta^{(1)}_{C_{2257}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{24MF}$

Quantity: $2, 2, 5, 7$

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\, ye\overline{y_1 y_2 ye}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}\, ye\overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{2258}}$

Quantity: $2, 2, 5, 8$

$$16\pi^2\,\mu\,\frac{dC_{2258}}{d\mu} = \beta^{(1)}_{C_{2258}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{24MF}$

Quantity: 2, 2, 5, 8

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{MF}$
T2	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} ig_2^2\overline{y_1 y_2}}{3MF}$

$\beta^{(1)}_{C_{22511}}$

Quantity: $2, 2, 5, 11$

$$16\pi^2\,\mu\,\frac{dC_{22511}}{d\mu} = \beta^{(1)}_{C_{22511}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1\,y_2}}{M\,F}$
T2	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1\,y_2}}{M\,F}$
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1\,y_2}}{M\,F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 2, 2, 5, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}iye\overline{y_1 y_2 ye}}{12MF}$

$\beta^{(1)}_{C_{22512}}$

Quantity: 2, 2, 5, 12

$$16\pi^2\,\mu\,\frac{dC_{22512}}{d\mu} = \beta^{(1)}_{C_{22512}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$	$\frac{11\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{8MF}$

Quantity: 2, 2, 5, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$

$$\beta^{(1)}_{C_{2267}}$$

Quantity: $2, 2, 6, 7$

$$16\pi^2\,\mu\,\frac{dC_{2267}}{d\mu} = \beta^{(1)}_{C_{2267}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{24MF}$	$-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 2, 2, 6, 7

Terms 2 of 2

Column	Representative term structure
T1	$\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{M F}$
T2	$\frac{3^{\frac{3}{4}} i ye \overline{y_1 y_2 ye}}{M F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{\sqrt{2} \cdot 3^{\frac{3}{4}} i g_2^2 \overline{y_1 y_2}}{3 M F}$	$\frac{3^{\frac{3}{4}} i ye \overline{y_1 y_2 ye}}{12 M F}$

$\beta^{(1)}_{C_{2268}}$

Quantity: $2, 2, 6, 8$

$$16\pi^2\,\mu\frac{dC_{2268}}{d\mu}=\beta^{(1)}_{C_{2268}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1y_2}}{6MF}$	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1y_2}}{8MF}$	$\frac{17\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1y_2}}{24MF}$

Quantity: 2, 2, 6, 8

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\, ye\overline{y_1 y_2 ye}}{MF}$
T2	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3^{\frac{3}{4}}\, ye\overline{y_1 y_2 ye}}{12MF}$	$\frac{\sqrt{2}\cdot 3^{\frac{3}{4}}\, g_2^2\overline{y_1 y_2}}{3MF}$

$$\beta^{(1)}_{C_{22611}}$$

Quantity: 2, 2, 6, 11

$$16\pi^2\,\mu\,\frac{dC_{22611}}{d\mu} = \beta^{(1)}_{C_{22611}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}\,g_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\,lambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,g_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 2, 2, 6, 11

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} ye\overline{y_1 y_2 ye}}{12MF}$

$\beta^{(1)}_{C_{22612}}$

Quantity: 2, 2, 6, 12

$$16\pi^2\,\mu\,\frac{dC_{22612}}{d\mu} = \beta^{(1)}_{C_{22612}}$$

Terms 1 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{MF}$
T2	$\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{MF}$
T3	$\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{11\cdot 3^{\frac{3}{4}}\,ig_2^2\overline{y_1 y_2}}{8MF}$	$-\frac{3^{\frac{3}{4}}\,ilambda12Inv1\overline{y_1 y_2}}{6MF}$	$-\frac{3^{\frac{3}{4}}\,ig_1^2\overline{y_1 y_2}}{8MF}$

Quantity: 2, 2, 6, 12

Terms 2 of 2

Column	Representative term structure
T1	$\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{MF}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{3^{\frac{3}{4}} iye\overline{y_1 y_2 ye}}{12MF}$